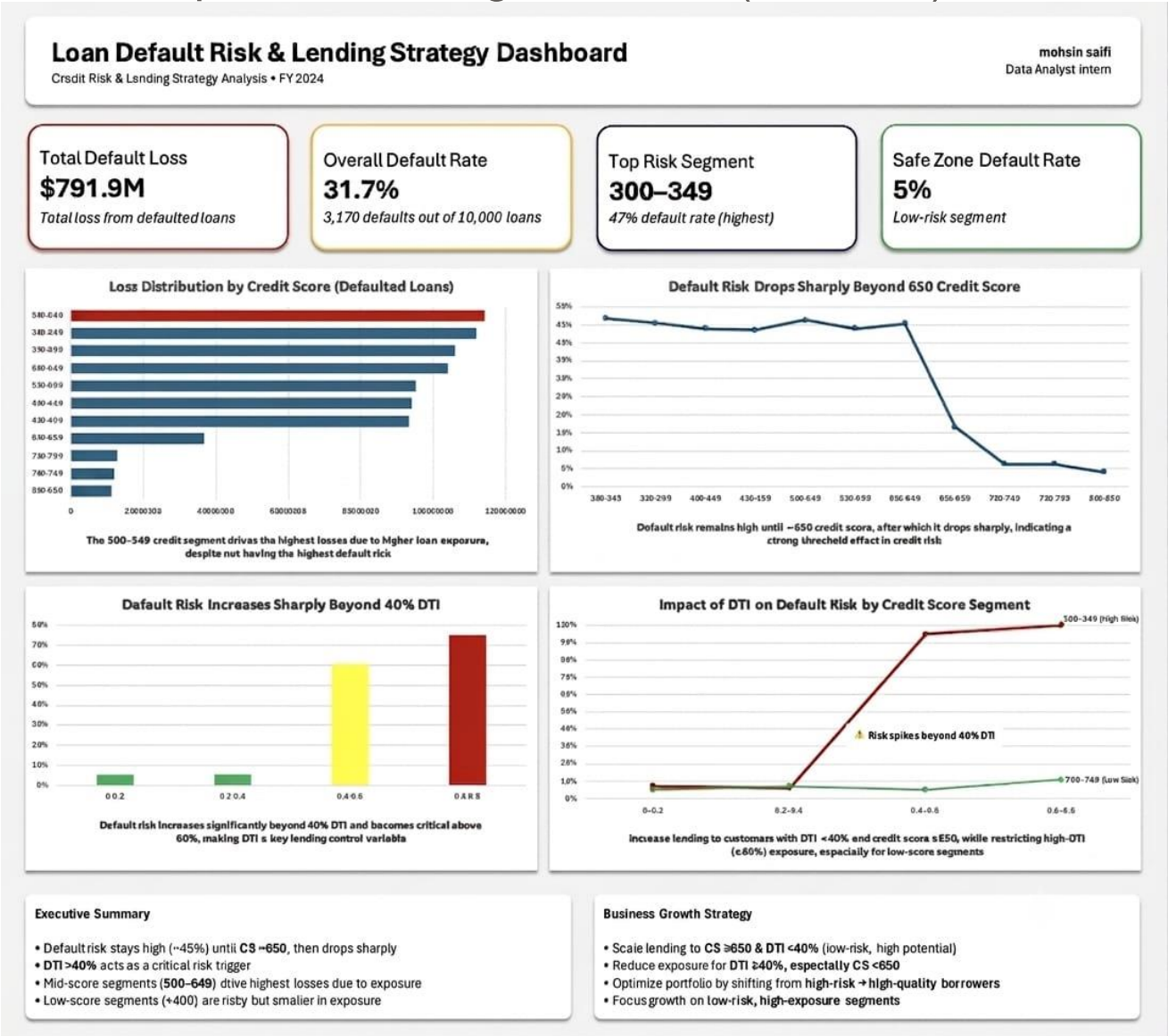


Loan Portfolio Risk & Lending Strategy Dashboard

Data-driven insights to reduce default risk and optimize lending decisions (FY 2024)



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Objective

Identify key risk drivers, quantify loss concentration, and recommend data-driven strategies to optimize lending decisions.

Business Questions

1. Where is the bank losing the most money?
2. Which customer segments are high-risk?
3. What factors drive loan defaults?
4. Where can the bank safely increase lending?

Data Overview

Dataset of 10,000 loan records including:

- Credit Score (banded)
- Debt-to-Income Ratio (DTI)
- Loan Amount • Default Flag (0/1)

Derived features:

- Credit Score Bands
- DTI Bands
- Default Rate (%)

Key Insights

1. Loss Concentration
 - Highest losses in 500–549 segment (~\$114M)
 - Driven by high exposure, not highest risk
2. Credit Risk Threshold
 - Default rate ~45% until CS ~650
 - Drops sharply beyond this point
3. DTI as a Risk Driver
 - Low risk for DTI <40%
 - Sharp increase beyond 40%, extreme above 60%

4. Interaction Effect (CS × DTI)

- Low-score borrowers highly sensitive to DTI
- High-score borrowers remain relatively stable

Business Impact

- Identified major loss concentration in mid-credit segments
- Highlighted high-risk exposure areas affecting portfolio quality
- Established clear thresholds (CS 650, DTI 40%) for risk control

Recommendations

- Scale lending to CS ≥ 650 & DTI $< 40\%$
- Restrict exposure for DTI $\geq 40\%$, especially CS < 650
- Rebalance portfolio toward low-risk, high-quality borrowers
- Monitor high-risk segments with high exposure

Segment Strategy

Segment	Action
High Risk + High Exposure	Reduce
High Risk + Low Exposure	Monitor
Low Risk + High Exposure	Scale
Low Risk + Low Exposure	Maintain

Conclusion

Financial losses are driven more by exposure concentration than default probability alone.

Combining credit score thresholds and DTI controls enables both risk reduction and strategic growth.

Tools Used

- Microsoft Excel (Pivot Tables, Dashboarding)
- Data Visualization
- Risk Segmentation Analysis

Project Outcome

Developed a decision-focused dashboard that:

- Identifies key risk drivers
- Highlights loss concentration
- Provides actionable lending strategy

Key Takeaway

Effective lending requires balancing risk (default rate) and exposure (loan volume) while considering how multiple factors interact.